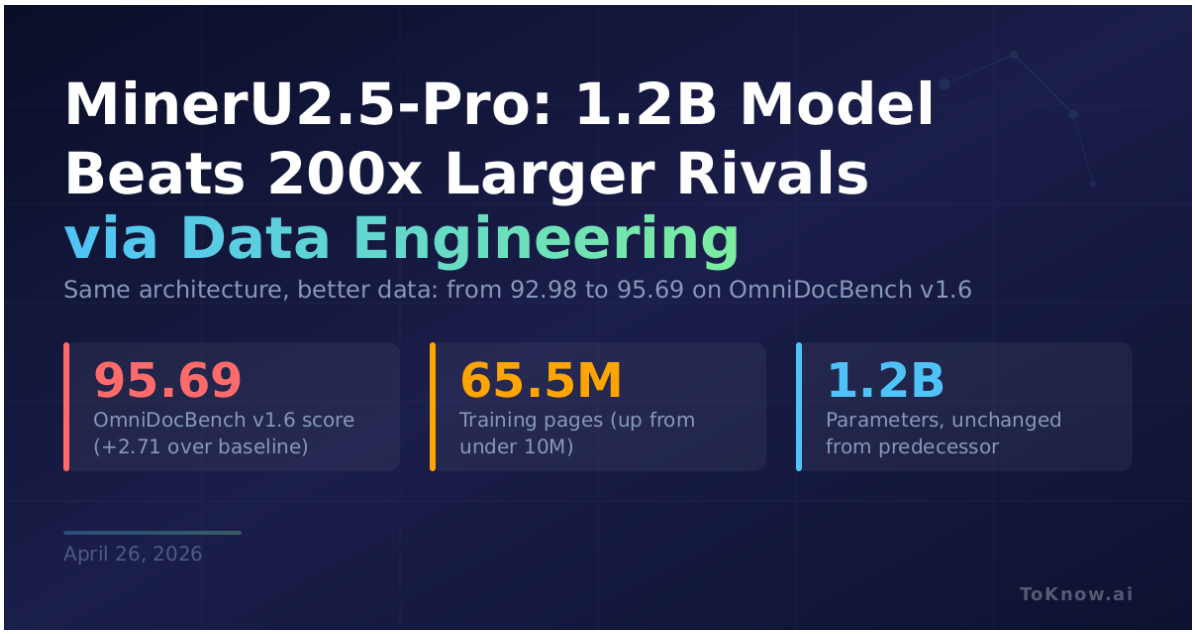
MinerU2.5-Pro: A 1.2B Model Beats 200x Larger Rivals by Fixing the Data, Not the Architecture

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2026-04-26

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OpenDataLab released MinerU2.5-Pro, a document parsing model that converts PDFs into structured Markdown. It keeps the exact same 1.2B-parameter architecture as its predecessor and improves the overall score on OmniDocBench v1.6 from 92.98 to 95.69, beating every

other system tested, including Gemini 3 Pro, Qwen3-VL-235B, and specialized OCR models with 200x more parameters. All gains came from a three-component Data Engine. Diversity-and-Difficulty-Aware Sampling expanded training data from under 10M to 65.5M pages while correcting long-tail distribution bias. Cross-Model Consistency Verification ran three different models on each sample and used their agreement to classify difficulty and generate reliable annotations. A Judge-and-Refine pipeline rendered LaTeX and HTML outputs back into images, then visually compared them to the originals to catch structural errors that text-only checks miss. A three-stage training progression (large-scale pretraining, hard-sample fine-tuning, GRPO alignment) matched data quality tiers to training phases.

This gives you production-grade parsing of tables, formulas, charts, and multi-column layouts from a model that runs on 4GB of VRAM. For legal, financial, and medical pipelines processing millions of pages, that means enterprise-quality extraction without enterprise-scale GPUs. MinerU now supports PDF, DOCX, PPTX, and XLSX inputs and recently moved to an Apache 2.0-based license, removing a major barrier to commercial adoption.

State-of-the-art models across wildly different architectures fail on the same hard samples, which points to training data as the shared bottleneck. Better annotation pipelines, not bigger models, are the lever. MinerU-Diffusion, from the same team, recently attacked decoding speed; MinerU2.5-Pro attacks data quality. Together they suggest document parsing is entering a phase where infrastructure matters more than architecture.

Sources:

- MinerU2.5-Pro Paper (arXiv)
- MinerU2.5-Pro Model (HuggingFace)
- MinerU GitHub Repository
- OmniDocBench Evaluation Framework

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